

CSL4030 – Data Engineering Project Proposal

Real-Time Financial Data Intelligence Platform

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Problem Statement

Individuals increasingly manage their finances across multiple sources such as bank accounts, UPI transactions, credit cards, and digital wallets. Although financial aggregation and expense-tracking applications exist, transaction data remains heterogeneous, fragmented, and difficult to transform into reliable, personalized, real-time insights.

The challenge is not simply collecting financial transactions, but building a scalable data platform that can ingest data from different formats and sources, preserve the original data, clean and normalize transactions, detect data-quality issues and duplicates, and make the resulting information available for both batch analytics and real-time processing.

Our Project aims to address this problem by building an end-to-end financial data intelligence platform. The system will ingest realistic financial transaction data from heterogeneous sources, store raw data in a data lake, transform and model the data using ETL pipelines, and maintain a structured analytical warehouse.

Also, our project will simulate real-time transaction streams using Kafka and process them using Spark Structured Streaming. A machine-learning-based anomaly detection model will learn historical spending patterns and identify transactions that are unusual for an individual user. The platform will also detect recurring expenses and generate personalized spending insights.

The processed data will be exposed through an interactive dashboard showing spending trends, category-wise expenses, recurring payments, anomalies, and other financial indicators. A separate monitoring layer will track pipeline health, data freshness, processing latency, failures, and streaming lag.

Core Objective of the Project

The objective of our project is to demonstrate how heterogeneous financial transaction data can be transformed into reliable, real-time, personalized financial intelligence through a scalable data engineering along with a machine learning architecture.

What we expect to solve:

- Where is the user's money being spent?
- How has spending changed over time?
- Which expenses are recurring?
- Which transactions are unusual compared with the user's historical behavior?
- Which spending categories are increasing unusually fast?
- Can the system process new transactions in near real time?
- Can the complete data pipeline be monitored for quality, freshness, and reliability?

Project Categories:

- ETL/ELT Data Pipelines (Mostly included)
- Real-time Stream Processing Architectures (Mostly included)
- Data Warehouse Architectures (Mostly included)
- Lake Architecture (Mostly included)
- Cloud-Native Architectures (slightly included)
- Microservices Architectures (slightly included)
- Analytics & Monitoring Systems (slightly included)

Expected Outcome

Our proposed pipeline (Might change as and when we build the system):

Heterogeneous Financial Data → Data Lake → ETL/ELT → Data Quality → Data Warehouse → Real-Time Streaming → ML Anomaly Detection → Financial Insights → Dashboard & Monitoring

Therefore, our project will not be another expense-tracking application, but as a financial data intelligence platform demonstrating modern data engineering, streaming and analytics.

Project Tracker can be found here:

<https://docs.google.com/spreadsheets/d/1fpiK4V7PMoLZIQGeToU6TPPsOTOnqbKhtB8tnQUCrVc/edit?usp=sharing>